National University of Computer and Emerging Sciences, Lahore Campus

Course:	Parallel and Distributed Computing	Course Code:	CS-3006
Program:	BSCS & BSDS	Semester:	Spring 2024
Duration:	7 Days	Total Marks:	30
Submit Date:	02-Apr-2024	Weight	1.25%
Type:	Announced	Page(s):	9
Exam:	Assignment 03 Solution	Section:	

Name & Roll No:

Submission Mode & Time: Handwritten solutions on A4 papers to be submitted during the lecture. You must have to write all the steps involved in the solution. State your answers in the readable handwriting and with the help of diagrams, where necessary.

Question # 1: **[6 marks, CLO # 2]**

Provide the output of the given code considering compiler will generate maximum 4 threads if dynamic adjustment is enabled.

```
workers = omp_get_max_threads();          //can use num_procs

printf ("%d maximum allowed threads\n", workers);

printf ("Total number of allocated cores are:%d\n", omp_get_num_procs());

omp_set_dynamic(1);          // dynamic adjustment enabled

omp_set_num_threads(8);

printf ("Total number of requested when dynamic is true are:%d\n", 1);

#pragma omp parallel {

        #pragma omp single nowait

        printf("Total threads in parallel region1=%d:\n", omp_get_num_threads());

        #pragma omp for

        for (i = 0; i < mult; i++)

        {a = complex_func();}

 }

omp_set_dynamic(0);          // dynamic adjustment disabled
```

```
omp_set_num_threads(6);

printf("Total number of requested when dynamic is false are:%d\n", 6);

#pragma omp parallel

{

#pragma omp single nowait

printf("Total threads in parallel region2=%d:\n", omp_get_num_threads());

#pragma omp for

for (i = 0; i < mult; i++)

{a = complex_func();}

}
```

Output:

4 maximum allowed threads

Total number of allocated cores are: 4

Total number of requested when dynamic is true are: 4

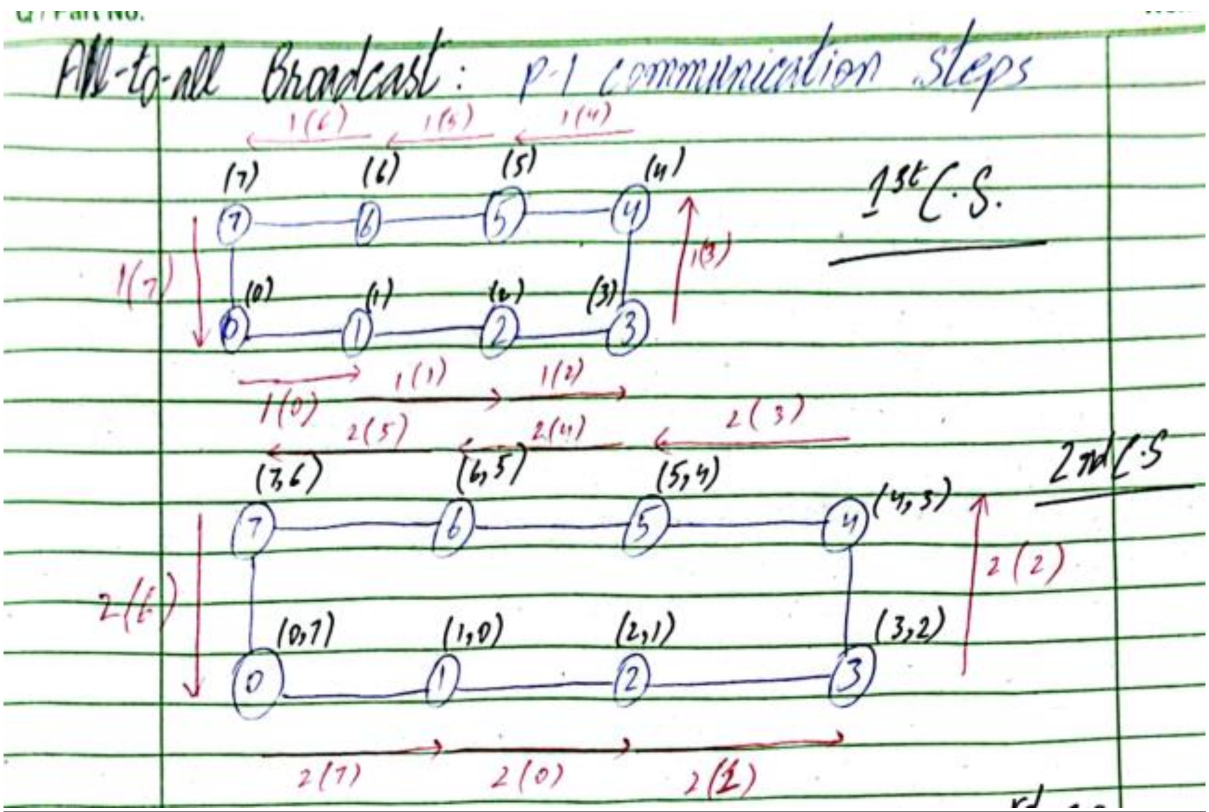
Total threads in parallel region1 = 4

Total number of requested when dynamic is false are: 6

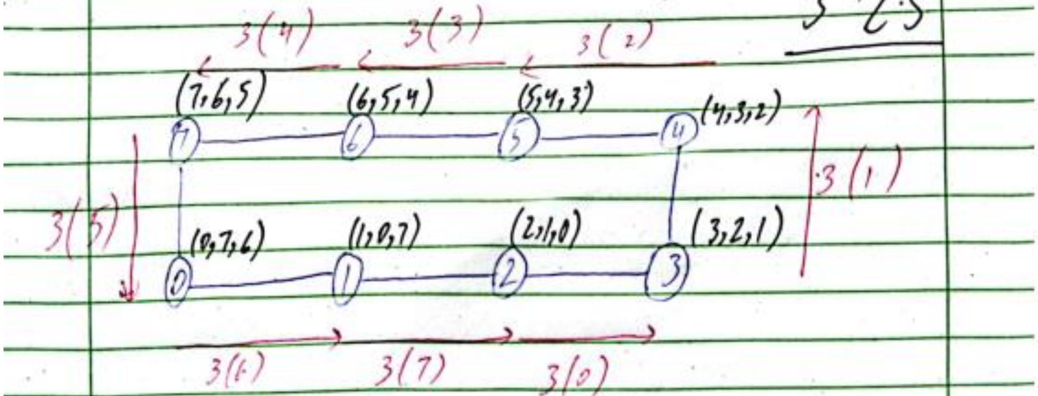
Total threads in parallel region2 = 6

Question # 2: **[8 marks, CLO # 3]**

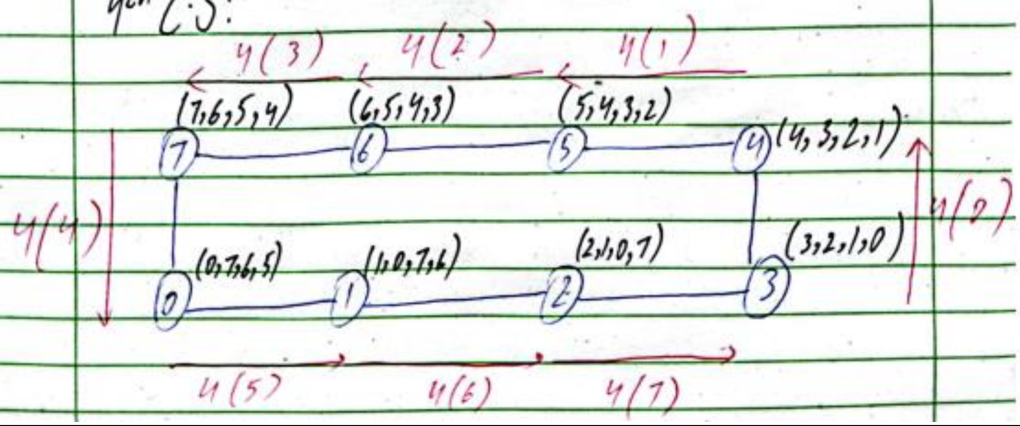
Perform all-to-all broadcast on 8 nodes ring structure. Write all the seven communication steps and the final state.



3rd C.S

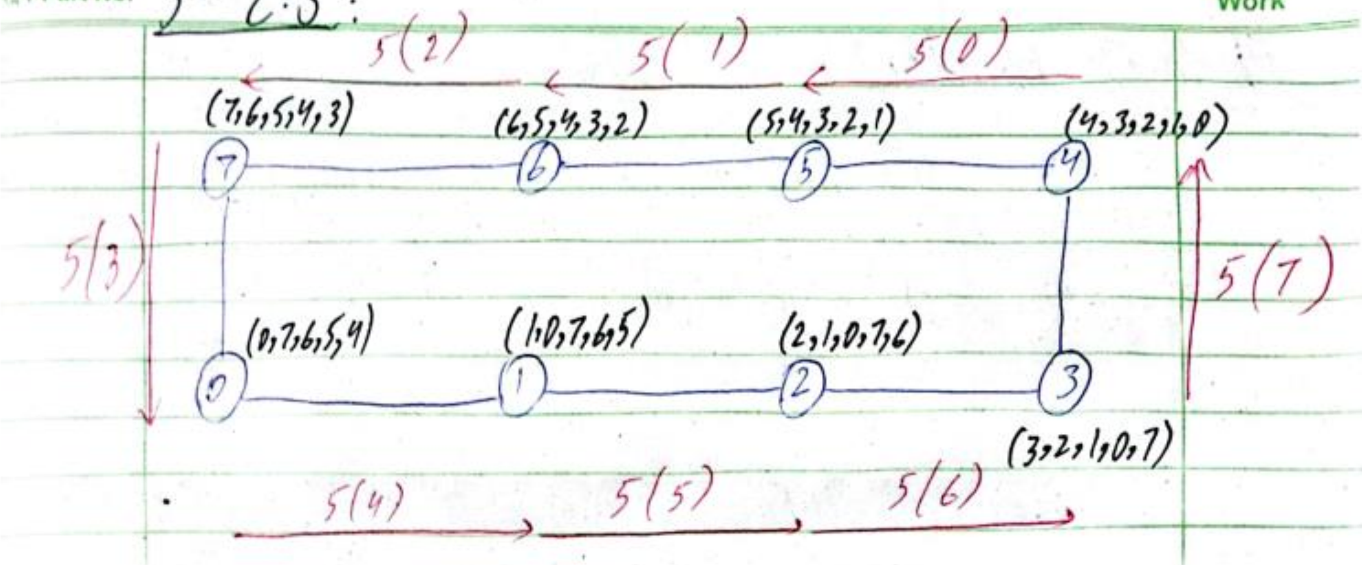


4th C.S:

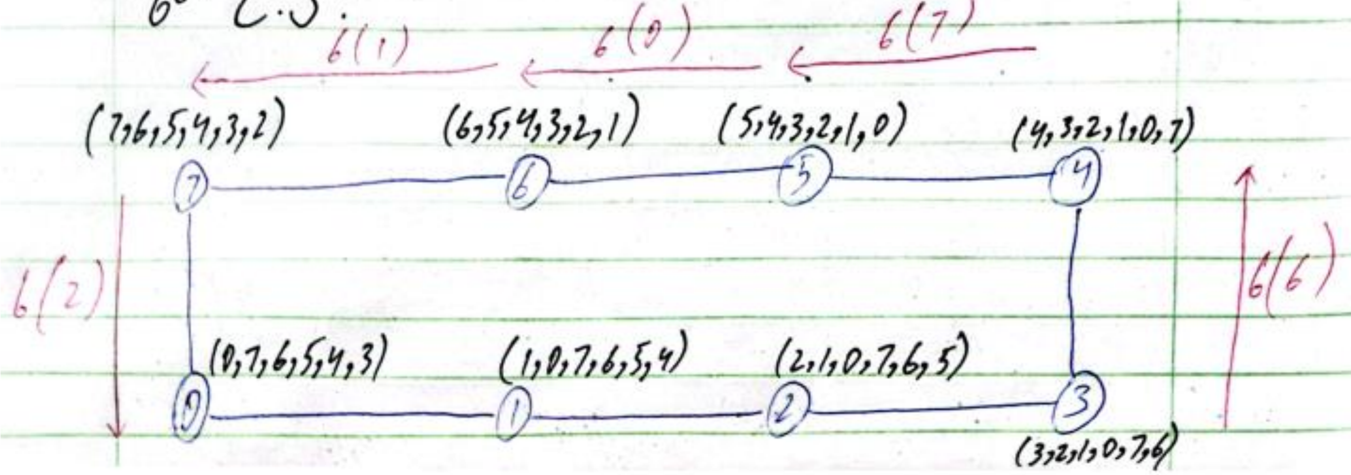


Q / Part No. 5th C.S:

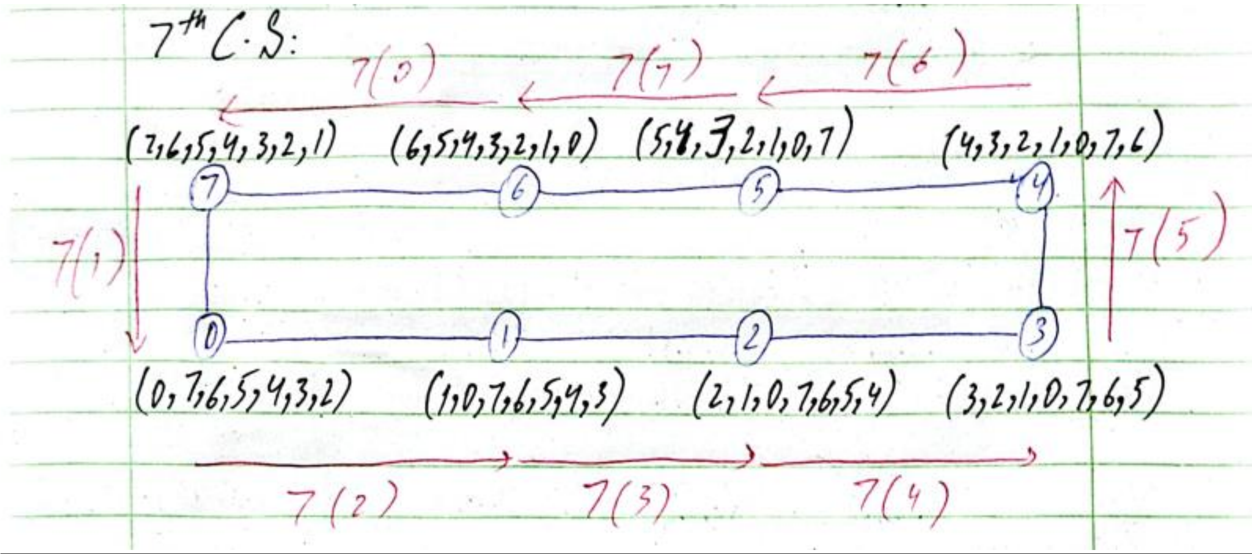
Rough Work



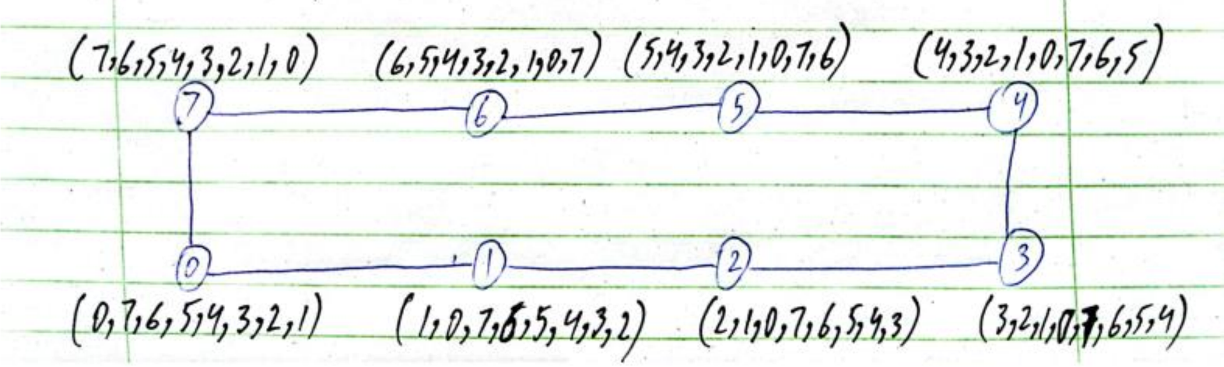
6th C.S:



7th C.S:



Result:



Question # 3: **[8 marks, CLO # 3]**

Perform prefix-sum operation on 3-D hypercube structure. Write all the steps involved and final state.

Prefix Sum:

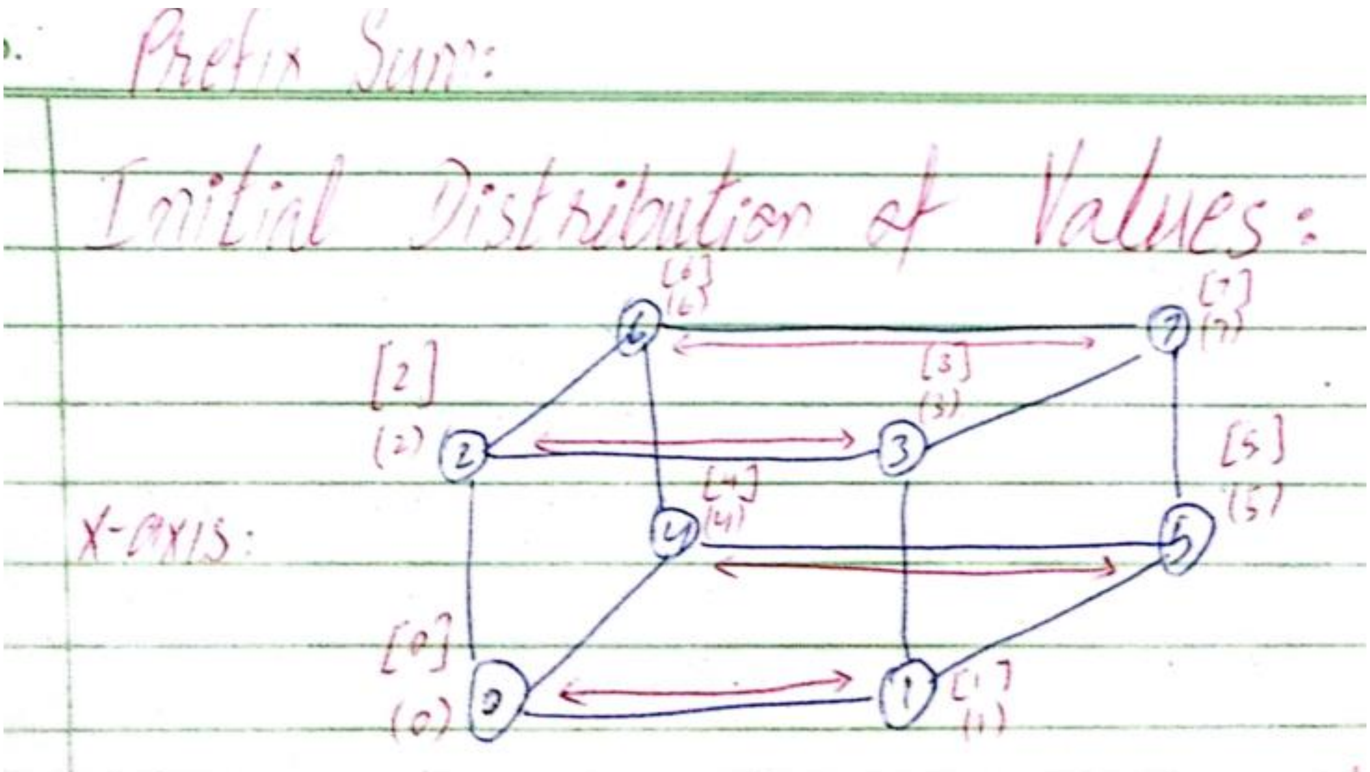
Round Bracket: That message is sent to other node in next step

Square Bracket: That message is kept with the node

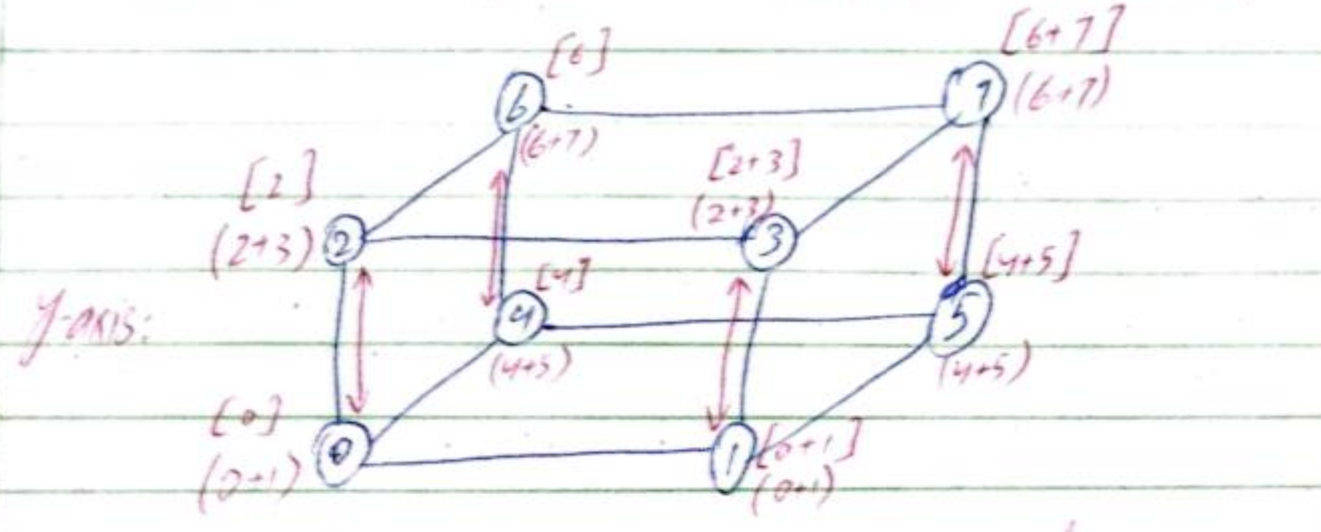
Lower index node will keep message in square bracket as it is.

Higher index node will add message in square bracket that it got from lower index node.

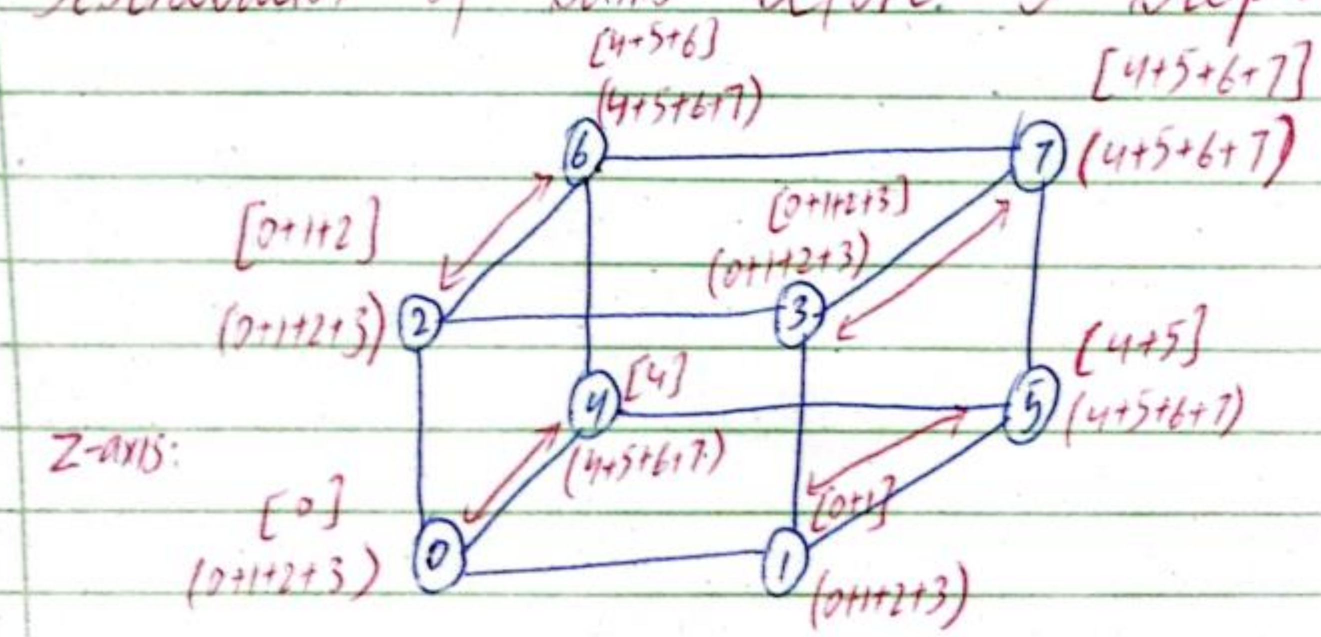
Communication cost is same as one-to-all broadcast



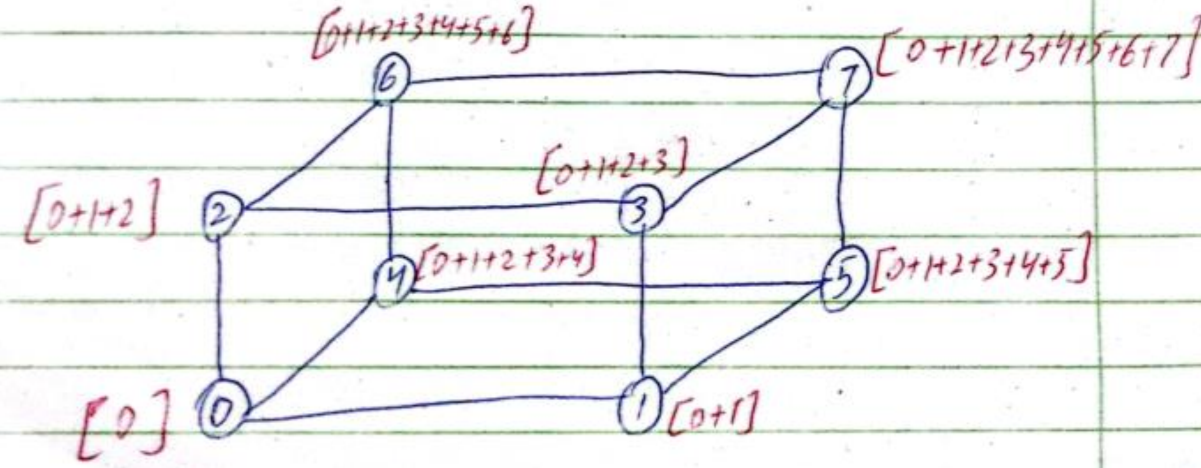
Distribution of Sums Before 2^{nd} Step:



Distribution of Sums Before 3^{rd} Step:



Final Distribution of Prefix Sums:



Question # 4: **[2 + 3 + 3 marks, CLO # 3]**

Assume we have a 4*4 matrix with the following values:

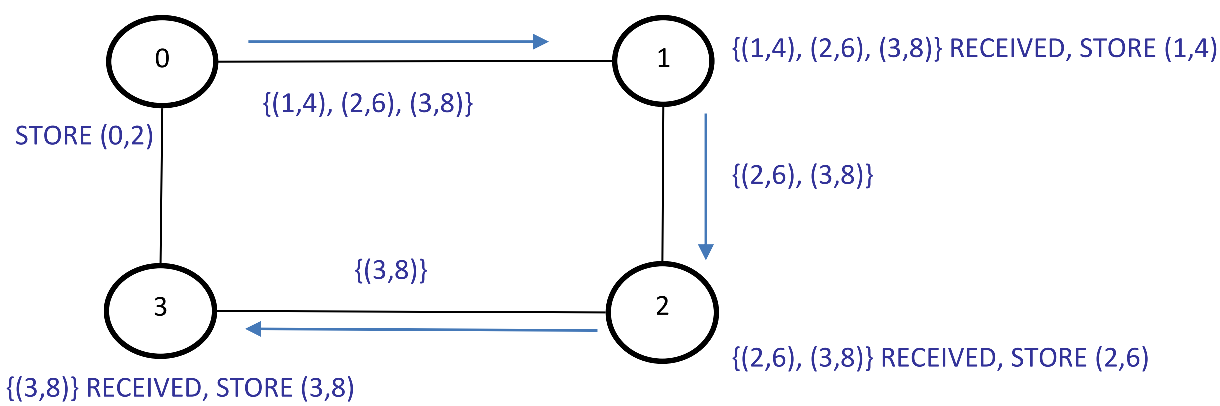
2 4 6 8
1 3 5 7
10 12 14 16
11 13 15 17

Assume each row is stored at different processes, row 1 (2, 4, 6, 8) is stored at process P0, row 2 at P1, row 3 at P2, and row 4 (11, 13, 15, 17) at P3. We want to apply a matrix transpose. Describe:

(i) The operation that needs to take place

Total exchange (a form of all-to-all personalized communication)

(ii) Draw the message originating from process P0 and show what happens at each step with this message



(iii) Draw the message originating from process P3 and show what happens at each step with this message